

Methicillin-Resistant *Staphylococcus aureus* (MRSA) Strain ST398 Is Present in Midwestern U.S. Swine and Swine Workers



Background Recent research has demonstrated that many swine and swine farmers in the Netherlands and Canada are colonized with MRSA. However, no studies to date have investigated carriage of MRSA among swine and swine farmers in the United States (U.S.). **Methods** We sampled the nares of 299 swine and 20 workers from two different production systems in Iowa and Illinois, comprising approximately 87,000 live animals. MRSA isolates were typed by pulsed field gel electrophoresis (PFGE) using *Sma*I and *Eag*I restriction enzymes, and by multi locus sequence typing (MLST). PCR was used to determine *SCCmec* type and presence of the *pvl* gene. **Results** In this pilot study, overall MRSA prevalence in swine was 49% (147/299) and 45% (9/20) in workers. The prevalence of MRSA carriage among production system A's swine varied by age, ranging from 36% (11/30) in adult swine to 100% (60/60) of animals aged 9 and 12 weeks. The prevalence among production system A's workers was 64% (9/14). MRSA was not isolated from production system B's swine or workers. Isolates examined were not typeable by PFGE when *Sma*I was used, but digestion with *Eag*I revealed that the isolates were clonal and were not related to common human types in Iowa (USA100, USA300, and USA400). MLST documented that the isolates were ST398. **Conclusions** These results show that colonization of swine by MRSA was very common on one swine production system in the midwestern U.S., suggesting that agricultural animals could become an important reservoir for this bacterium. MRSA strain ST398 was the only strain documented on this farm. Further studies are examining carriage rates on additional farms.